Department of Applied Mathematics and Statistics
COLORADO SCHOOL OF MINES
Math 550: Numerical Solutions of Partial Differential Equations
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Numerical PDEs Homework #0
Due Tuesday, September 15, 2026

Problem 1: Finite differences Write a matlab code that uses finite differences to compute the first derivative of

$$f(x) = \exp(\sin(x))$$

on the interval $0 \leq x \leq 2\pi$. Use second-order centered differences on internal grid points and first-order one-sided differences at endpoints.

(a) Show a graph of the exact analytic solution $f'(x)$ and your numerical solution $f'_n(x)$.

(b) Show a relative error convergence plot by plotting the relative error on the y-axis and the grid size N on the x-axis. Measure relative error as

$$Err = \frac{\|f'(x) - f'_n(x)\|_p}{\|f'(x)\|_p}.$$

Plot the error using both the $p = \infty, 2$ norms in a log-log scale. The relative errors should be lines with slopes 1 and 3/2 respectively, i.e. it scales with $\frac{1}{N}, \frac{1}{N^{3/2}}$. Extra Credit: The 3/2 slope is strange, why is that happening?

Result:

(a)

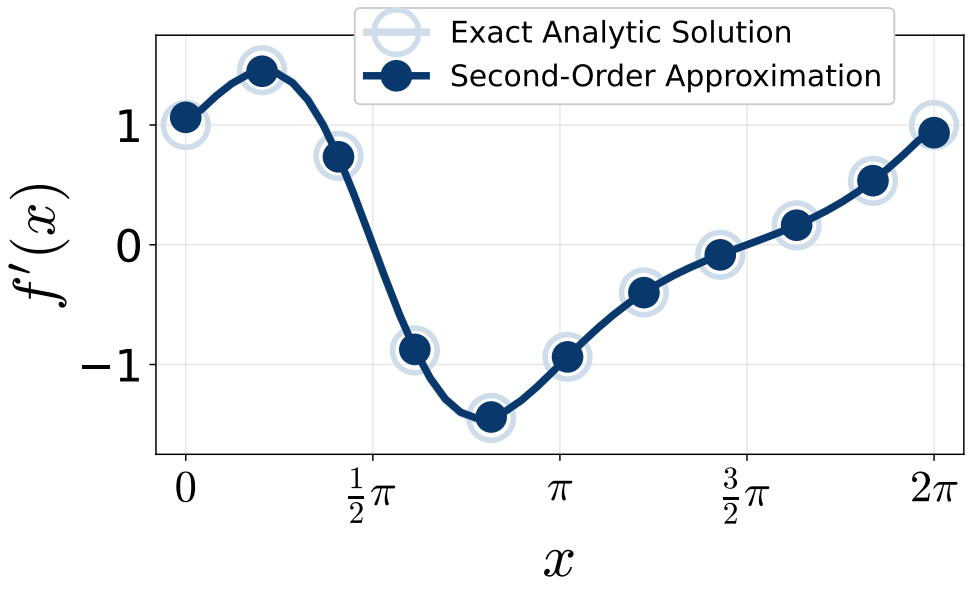


Figure 1.1: Accuracy of Second-Order approximation of the given function's derivative with $N = 50$.

(b)

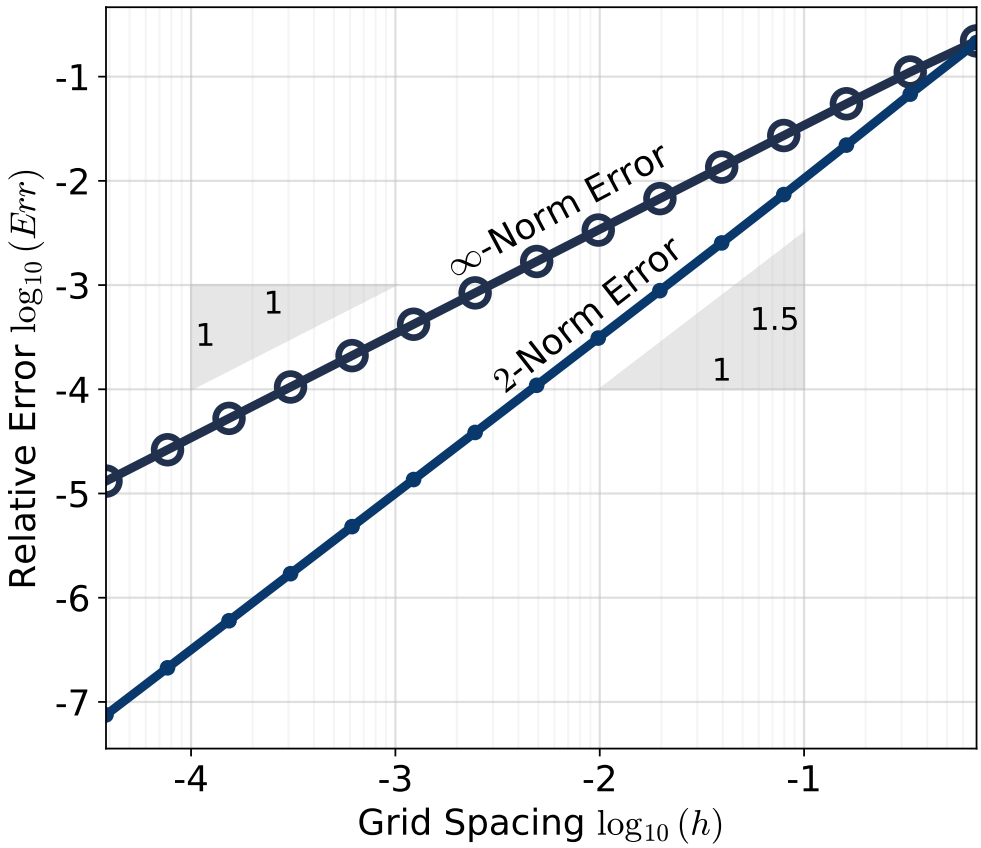


Figure 1.2: p-norm convergence for $p \in \{2, \infty\}$ of the derivative approximation.

Proof. We begin with the given equation for the 2-norm error:

$$Err_2 = \frac{\|f'(x) - f'_n(x)\|_2}{\|f'(x)\|_2}$$

When order notation is introduced, the 3/2 slope property appears. To begin, state that for all x_i, its derivative does not depend on h:

$$f'(x_i) \to O(1), \quad i \in \{1, 2, \ldots, N\}$$

Next, consider the error of the first-order-evaluated endpoints, for which it is well known their error scales with h:

$$|f'(x_i) - f'_n(x_i)| \to O(h), \quad i = 1, N$$

Similarly, the error for the second-order-evaluated interior points:

$$|f'(x_i) - f'_n(x_i)| \to O(h^2), \quad i \in \{2, 3, \ldots, N-1\}$$

Therefore the 2-norm of the error at x_i and the analytic solution at x_i can be calculated:

$$\|f'(x) - f'_n(x)\|_2 = \left(\sum_{i=1}^{N} f'(x_i) - f'_n(x_i)^2\right)^{1/2} \to \sqrt{O(h)^2 + O(N) \cdot O(h^2)^2},$$

where $O(h)$ comes from the two endpoints' error. Following simplifications:

$$\begin{aligned} O(h)^2 &= O(h^2), \\ O(N) \cdot O(h^2)^2 &= O(h^3) \quad (\text{by } N \to O(1/h)), \\ O(h^2) + O(h^3) &= O(h^2) \quad (\text{by } h \text{ small}), \\ \sqrt{O(h^2)} &= O(h) \end{aligned}$$

the order of the numerator term (that is, the norm of the error at every x_i) is yielded:

$$\|f'(x) - f'_n(x)\|_2 \to O(h)$$

A similar and more straightford process occurs for the denominator term (that is, the norm of the analytic solution at every x_i):

$$\|f'(x)\|_2 = \left(\sum_{i=1}^{N} f'(x_i)\right)^{1/2} \to \sqrt{O(N)} = O(h^{-1/2})$$

Therefore the order for the convergence emerges:

$$Err_2 \to \frac{O(h)}{O(h^{-1/2})} = O(h^{3/2}) = O(\frac{1}{N^{3/2}})$$

□

Problem 2: Second-order derivatives Repeat problem 1 but compute the second derivative $f''(x)$. Assume periodicity and wrap around the grid instead of using one-sided finite differences.

Result:

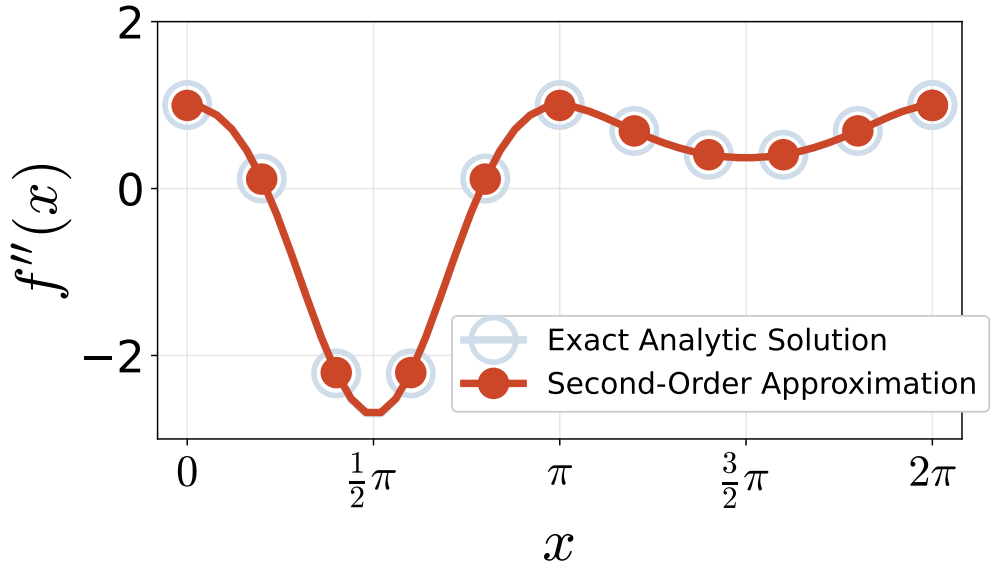


Figure 2.1: Accuracy of Second-Order approximation of the given function's second derivative with $N = 50$.

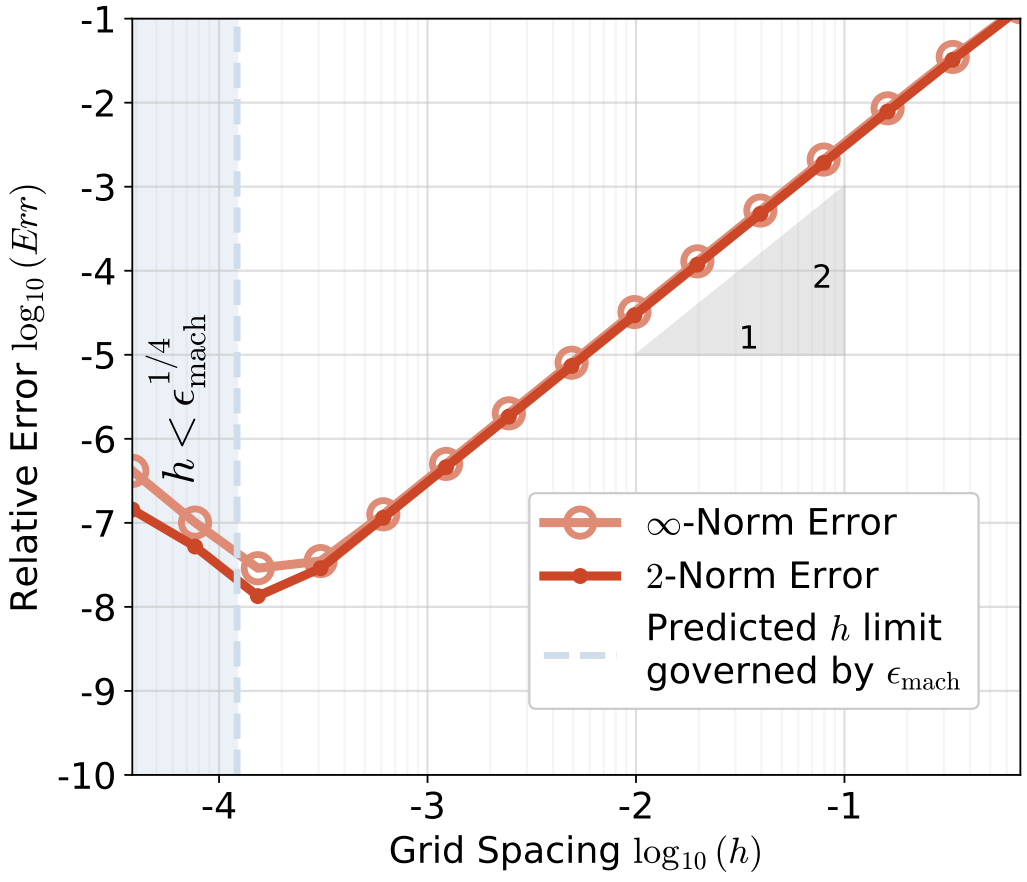


Figure 2.2: p-norm convergence for $p \in \{2, \infty\}$ of the second-derivative approximation.

Problem 3: ODE Consider the ODE

$$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + \sin(x)\frac{\mathrm{d}u}{\mathrm{d}x} + u(x) = f(x)$$

defined on $0 \leq x \leq 5$. Solve the ODE using second-order finite difference methods with the Dirichlet conditions. Use $u(x) = \sin(x)$ to test your solution by finding boundary conditions and the necessary forcing function $f(x)$. Show an error plot that proves your method converges with second order.

Result:

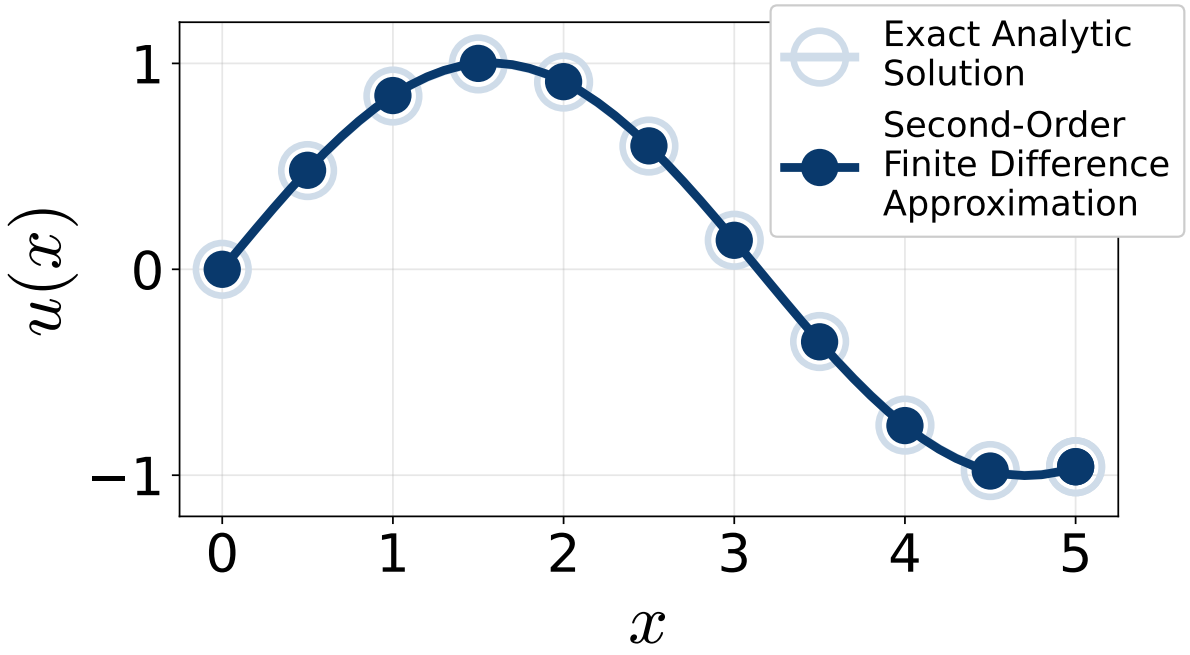


Figure 3.1: Accuracy of Second-Order Finite Difference Approximation of the test function $u = \sin(x)$ with $N = 50$.

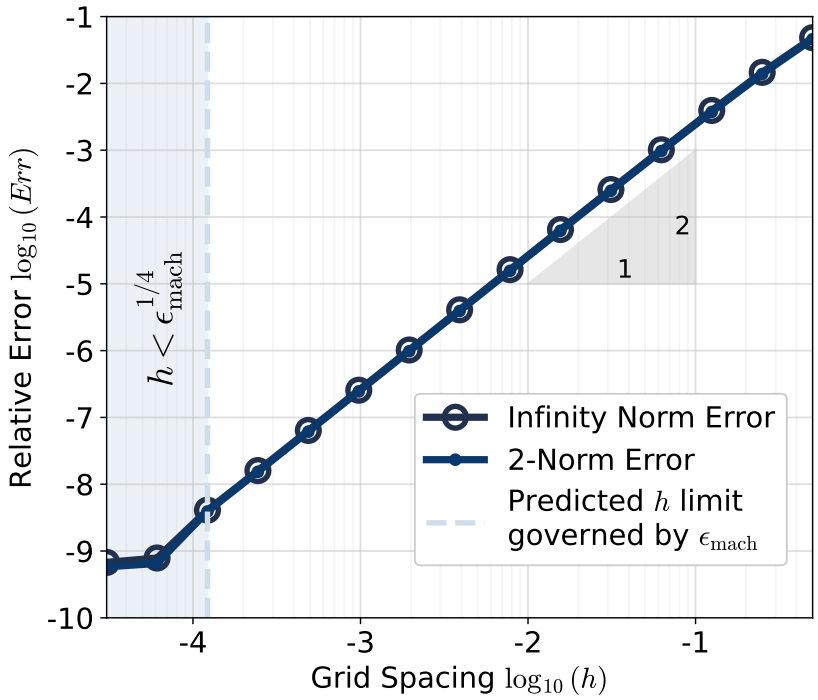


Figure 3.2: p-norm convergence for $p \in \{2, \infty\}$ of the finite difference approximation.

Problem 4: Poisson Equation Use finite difference methods to solve the Poisson equation $\nabla^2 u = f(x, y)$ in the domain $x \in [0, 5], y \in [0, 5]$. Use the test solution $u(x, y) = \sin(x)\cos(y)$.

(a) Apply Dirichlet conditions on all boundaries so that $u(x, y)$ satisfies the PDE and the boundary conditions, and show a plot of your solution and a second order convergence plot (similar to the first three problems).

(b) Modify your code to apply Neumann conditions on all boundaries. **This won't work.** Why not? – both in terms of the linear algebra and in terms of the actual PDE we want to solve. Suggest a solution to the issue.

Result:

(a)

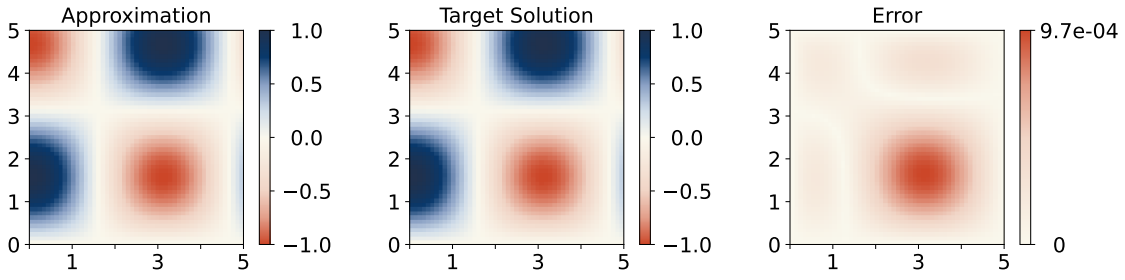


Figure 4.1: Accuracy of Second-Order Finite Difference Approximation of the test function $u = \sin(x)\cos(y)$ with $N = 50$. using known value for u at the boundaries.

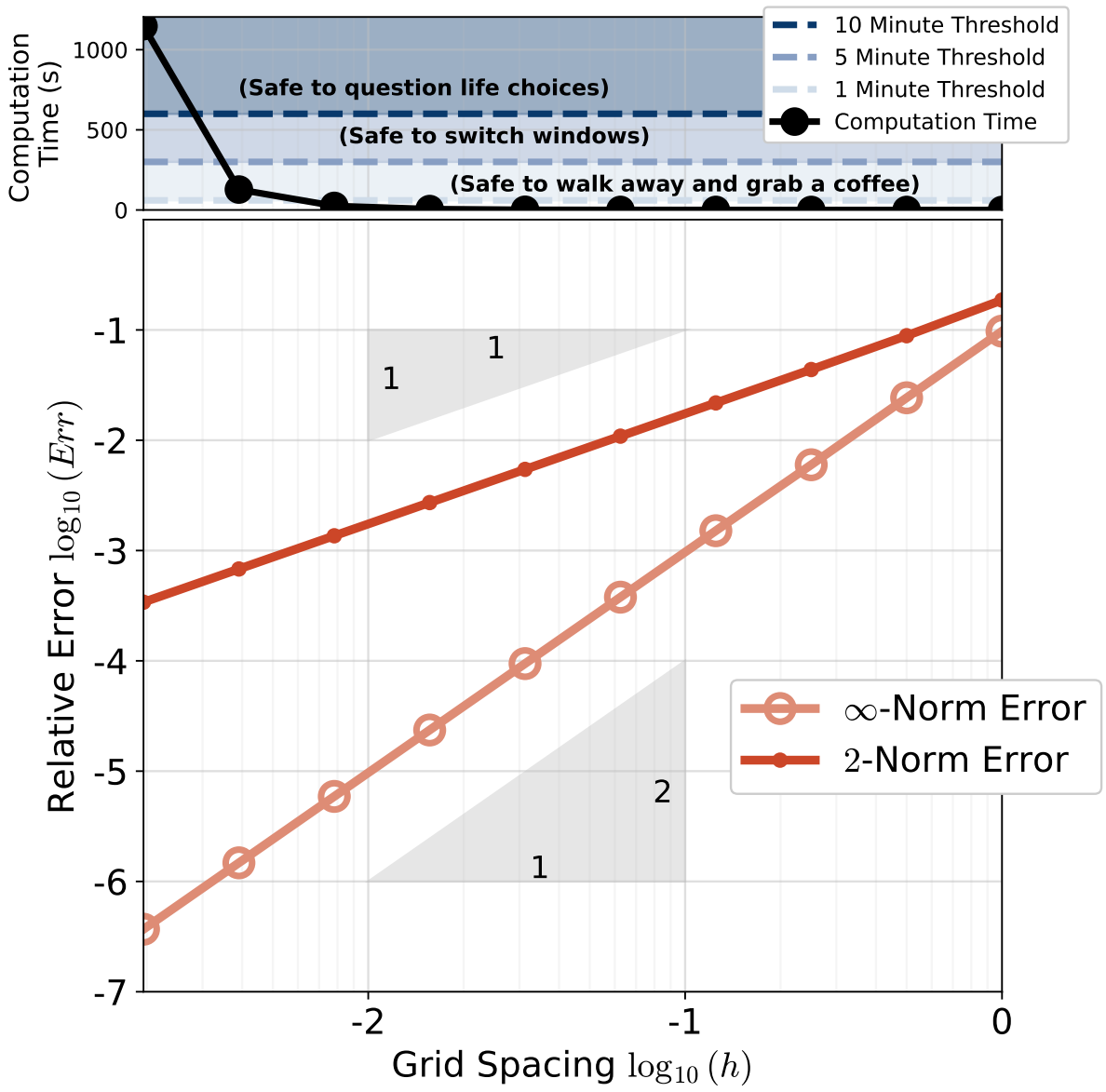


Figure 4.2: p-norm convergence for $p \in \{2, \infty\}$ of the Laplacian 5-point stencil approximation.

(b) *Proof.* Explanation of why Neumann fails:

The governing equations are only concerned with the first and second derivatives of u.

The fundamental theorem of calculus reminds us that integrating an undefined function gets us an arbitrary constant:

$$\int g(x)\,dx = \int g(x)\,dx + C$$

That tells us:

$$u \text{ solves some property based on } u_x,\, u_y \implies u + C \text{ solves the same property.}$$

Therefore, the linear equation for $u_{i,j}$ is *underdetermined* and will be correct, +/- a constant, C. Which, based on the solver used, can be close to the maximum magnitude the machine can compute. □

It can be, for example, accurate to give or take a constant of **2.3 100 billions.**

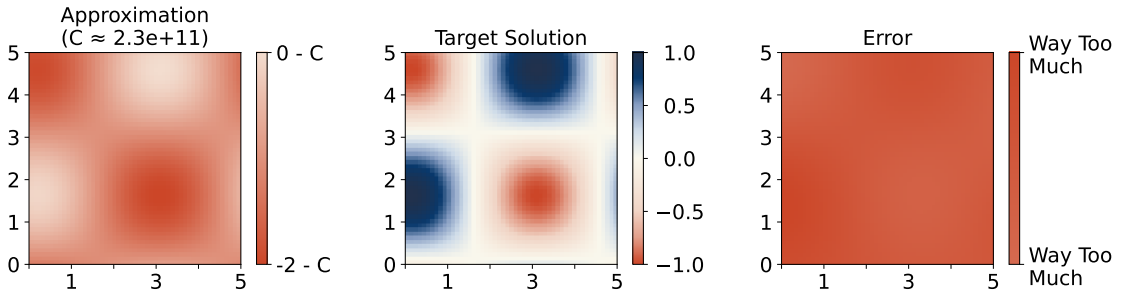


Figure 4.3: Accuracy of Second-Order Finite Difference Approximation of the test function $u = \sin(x)\cos(y)$ with $N = 50$. using known value for u_x, u_y at the boundaries.

However, let's say we know that this solution is correct to a constant, albeit a large one. If we also know what the minimum of the functin *should* be, then we can adjust accordingly:

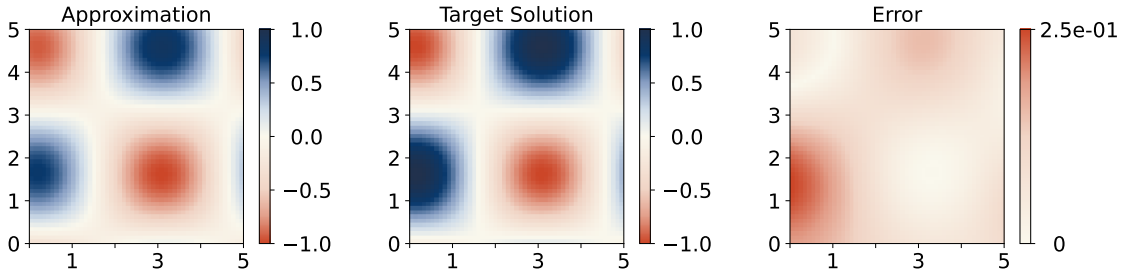


Figure 4.4: Accuracy of Second-Order Finite Difference Approximation of the test function $u = \sin(x)\cos(y)$ with $N = 50$. using known value for u_x, u_y at the boundaries and adjusting to a known minimum.

Alternatively, if we know a property like oscellacion about zero, we can enforce that the boundary's norm is zero (that is, $u(x, y)$ sums to zero on the boundary):

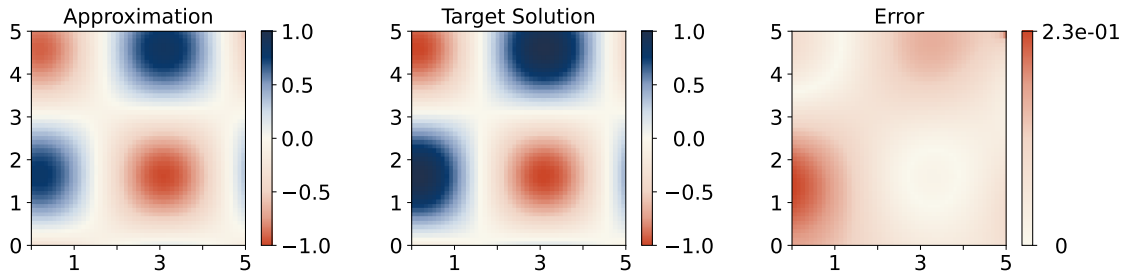


Figure 4.5: Accuracy of Second-Order Finite Difference Approximation of the test function $u = \sin(x)\cos(y)$ with $N = 50$. using known value for u_x, u_y at the boundaries and constraining the boundary norm to be zero.